

B.C.A. (Semester-II) Examination, 2020

C. PROGRAMMING AND DATA STRUCTURE

[Paper Code : BC-203]

Time : Three Hours

[Maximum Marks : 80]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. What is Data Structure? Discuss the types of data structure in details.
2. What is Recursive Function? Distinguish between Looping and Recursion with suitable examples.
3. What is Structure. Distinguish between structure and union. Write a program to demonstrate the concept of structure.
4. What is Linked List? Discuss the types of Linked List over Array.
5. What is Graph? Explain the properties of Graphs in details.
6. Write a 'C' program using function to sort the list of n integer elements into ascending order using bubble sort.
7. What is Stack? Write down the Algorithm for implementing push and pop operation in a Stack.
8. What is Tree? Discuss Binary Tree Traversal Recursive Algorithm in details.
9. What is Sorting? Discuss the types of sorting. Write an algorithm for selection sort for the list of n given numbers.
10. Write short notes on any two of the following:
 - (a) Application of Data Structure
 - (b) AVL Tree
 - (c) Dynamic Memory Allocation
 - (d) Tree Traversal

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